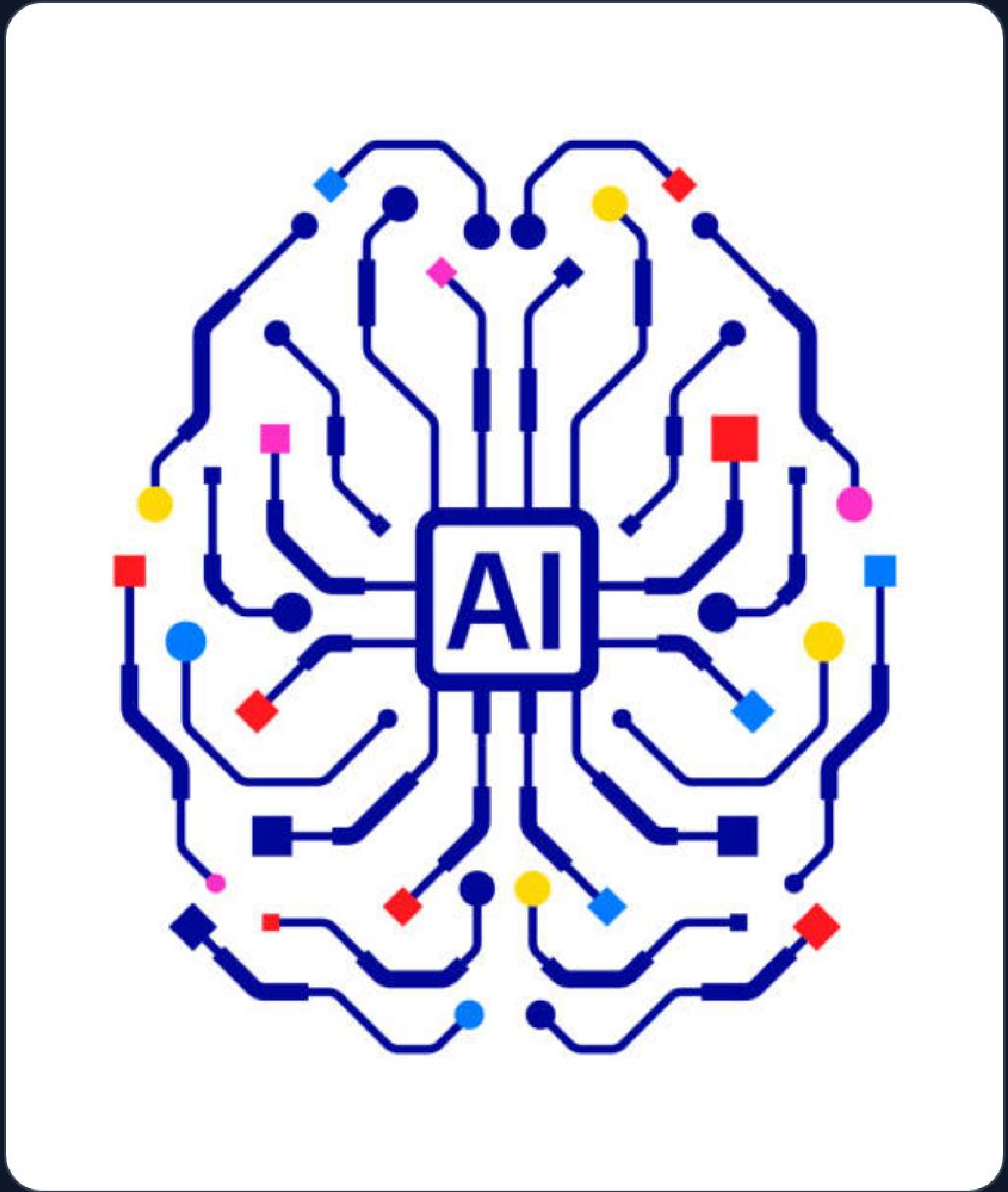
A PHILOSOPHICAL FRAMEWORK FOR LLMS

AI Under the Hood: Meaning, Use & Understanding

What would it take for an AI system to have true meaning, rather than merely produce meaningful-looking language?



“What would it take for an AI system to have *meaning*, rather than merely produce meaningful-looking language?”

Synthetic Generation

Genuine Intentionality

Three Philosophers, One Journey

A progressive investigation from mathematical structure to functional action

PHASE 01 • REPRESENTATION

Gottlob Frege

Focus: Sense & Reference. How does AI mathematically encode visual and semantic relationships?

PHASE 02 • USE

Ludwig Wittgenstein

Focus: Language Games. How does an AI model learn contextual usage from statistical patterns?

PHASE 03 • ACTION

John Searle

Focus: Speech Acts & Intent. Does appropriate linguistic behavior equate to real understanding?

MODULE 01

What is Meaning?

PHILOSOPHER

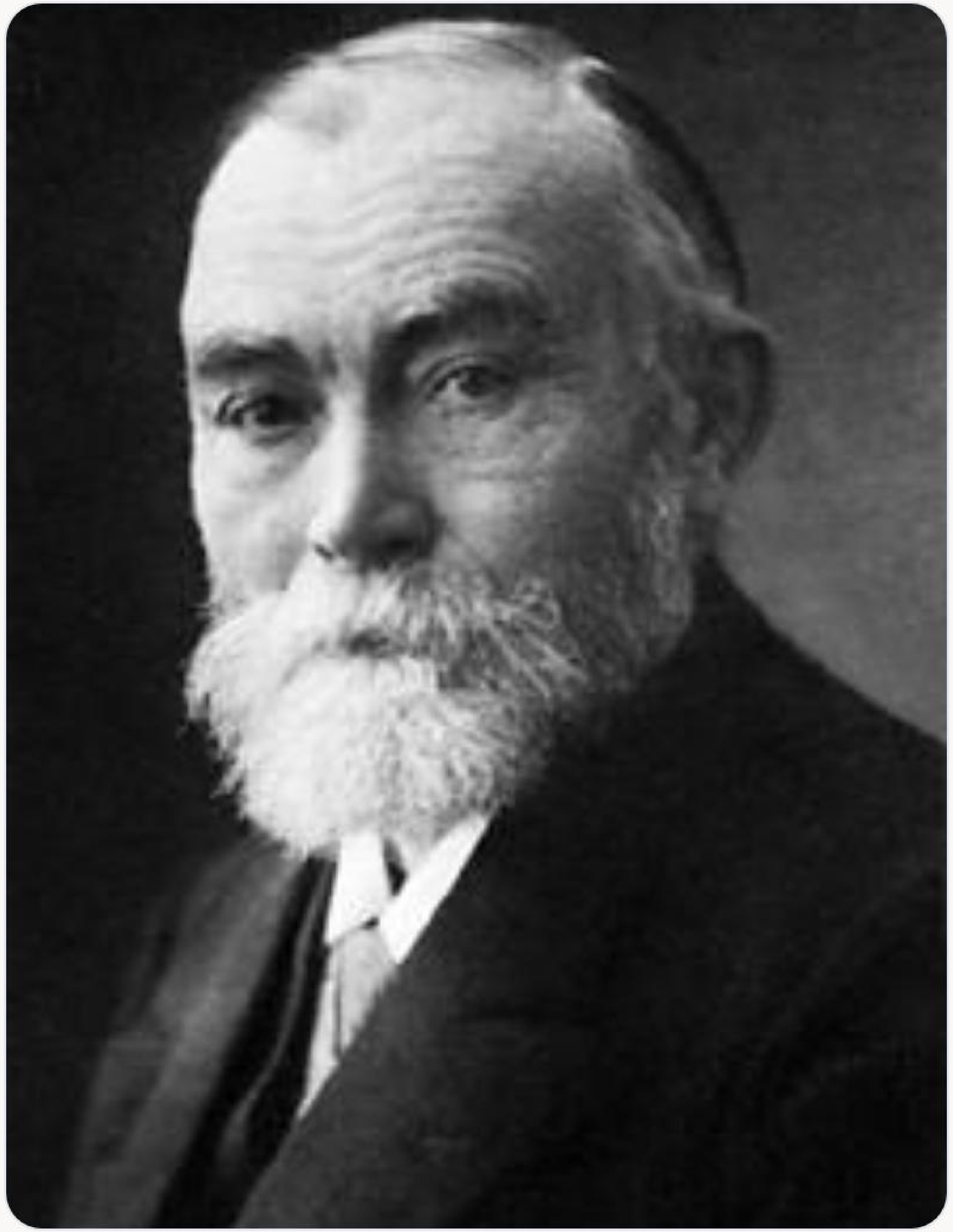
Gottlob Frege

AI CONCEPTS

Embeddings, High-Dimensional Vectors, Semantic Similarity

CORE QUESTION

"How can a machine represent meaning internally?"



Frege's Big Idea: Sense vs. Reference

Meaning requires distinguishing between the mode of presentation and the object itself

Sense 1 (Sinn)
"The Morning Star"

Sense 2 (Sinn)
"The Evening Star"

→

SHARED REFERENCE (BEDEUTUNG)

Planet Venus

The actual physical body in reality

Key Takeaway: Meaning is not just pointing to an object—it includes how the object is presented.

How AI Represents "Meaning"

Transforming linguistic tokens into high-dimensional vector space

Input Text
"Bank"

→

Token ID
[1842]

→

Dense Vector (Embedding)
`[-0.24, 0.81, 0.12, ...]`

Semantic Similarity

Words with similar meanings map to nearby geometric coordinates in vector space.

Contextual Representation

The vector for "Bank" shifts dynamically depending on whether it means *River Bank* or *Money Bank*.

FREGEAN SYNTHESIS

The Fregean Limit

Do vector embeddings amount to genuine meaning, or are they merely highly effective mathematical models for predicting text?

> **Class Discussion Case:** *"The morning star is the evening star."* An LLM calculates token probabilities accurately, but does it grasp **why** this statement yields new astronomical discovery?

Learning Objective: Representation is necessary for intelligence, but vector geometry alone does not settle the question of meaning.

MODULE 02

How Does Language Work?

PHILOSOPHER

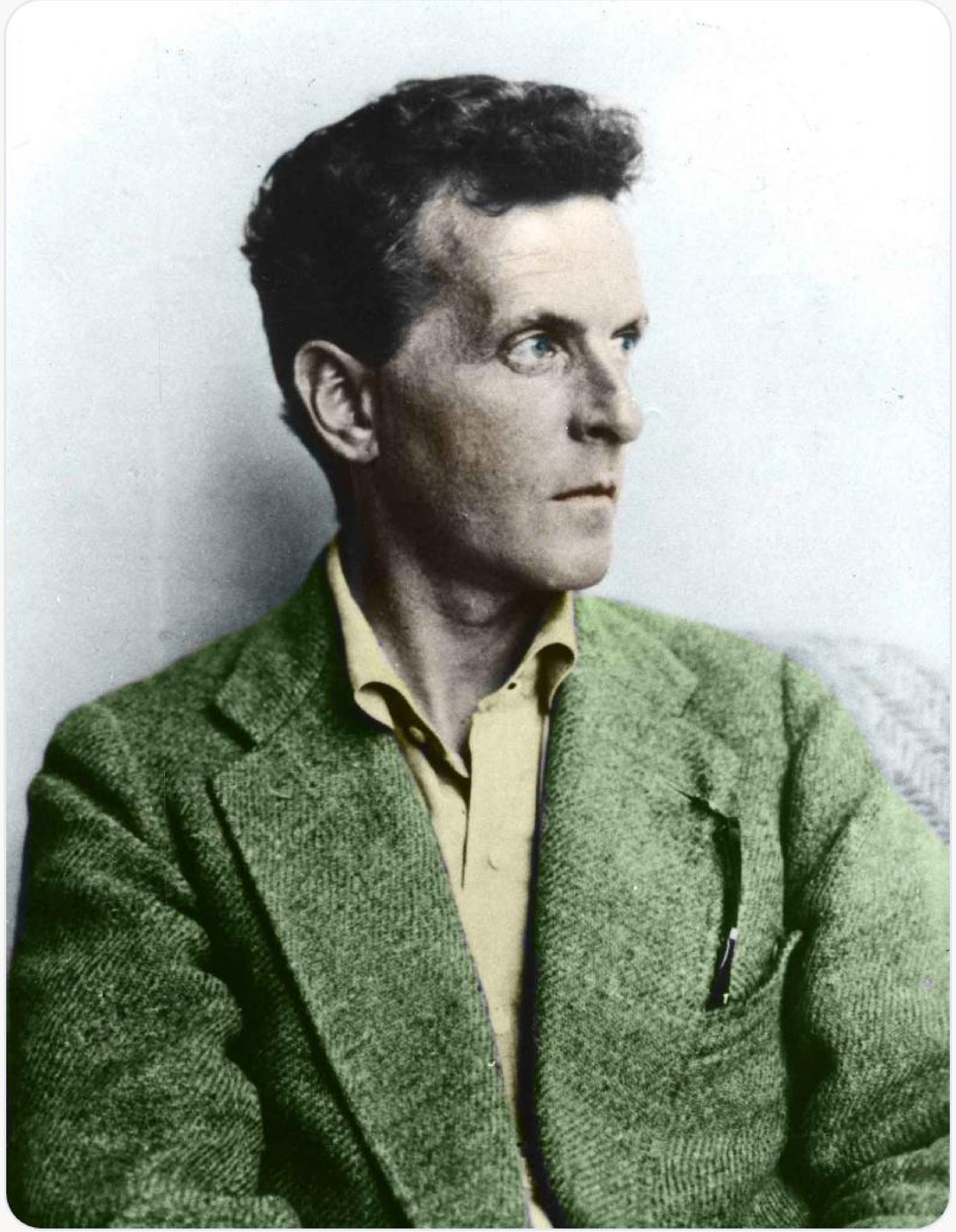
Ludwig Wittgenstein

AI CONCEPTS

Training Corpora, Context Windows, In-Context Learning

CORE QUESTION

"How does an AI model learn to use language appropriately in context?"



Wittgenstein’s Big Idea: Language Games

“The meaning of a word is its use in the language.” — Philosophical Investigations

Asking Questions

Seeking information or clarification

Giving Orders

Directing behavior and system action

Describing Objects

Mapping properties to reality

Telling Jokes

Subverting context and expectation

How AI Learns "Use"

Simulating linguistic games through statistical distribution and context adaptation

01

Pre-Training

Extracts statistical regularities across trillions of web tokens.

→

02

Next-Token Prediction

Calculates optimal continuation probabilities.

→

03

In-Context Learning

Adapts behavior dynamically based on system prompts.

WITTGENSTEINIAN CHALLENGE

Rule-Following vs. Pattern-Matching

Social Context Example:
“Can you pass the salt?”

Humans interpret this phrase as an indirect **action request**, not an enquiry into arm mobility. Is the AI following shared social "rules", or merely replicating high-frequency co-occurrence patterns?

Learning Objective: Language is not a rigid map of symbols; it is a shared social practice embedded in real-world context.

MODULE 03

What Are We Doing When We Speak?

PHILOSOPHER

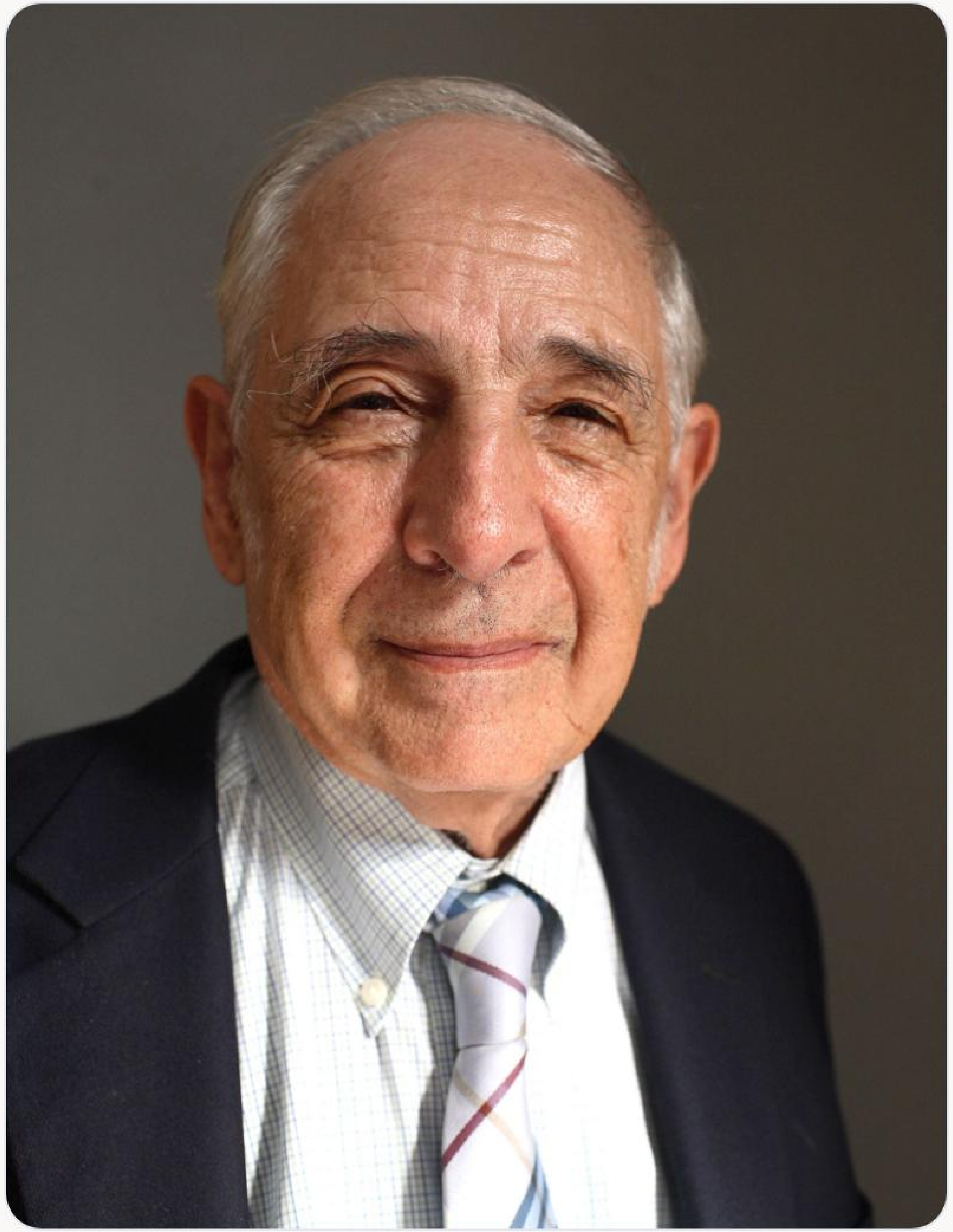
John Searle

AI CONCEPTS

Instruction-Following, Intent Classification, Tool Use & Agents

CORE QUESTION

"Does generating a linguistic response amount to performing a speech act?"



Searle’s Big Idea: Speech Acts

Language as action: Speaking is performing an intentional act with commitments

Asserting Facts
Committing to truth

Making Promises
Binding future action

Issuing Requests
Directing listener action

Apologizing
Expressing psychological state

The Essential Requirement: A genuine speech act requires underlying intention.

How AI Performs "Actions"

From text generation to execution within computational environments

Intent Classification

Categorizes user requests into query, command, or workflow execution.

Tool Execution

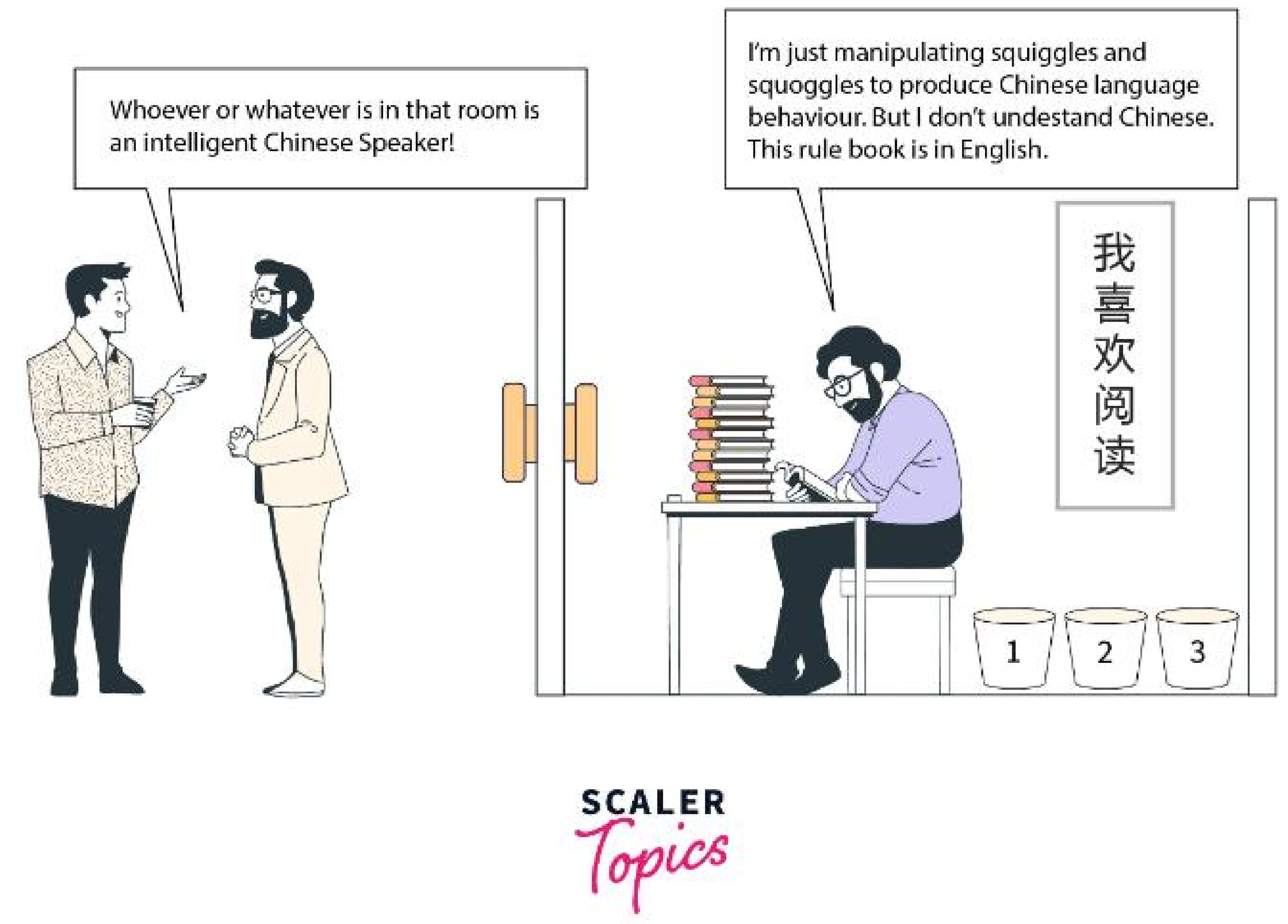
Invokes APIs, executes code blocks, and fetches external web content.

Autonomous Agents

Plans, evaluates feedback, and adapts actions to reach target objectives.

The Chinese Room Argument

Searle’s classic thought experiment on syntax versus semantics



01. Symbol Manipulation: A person inside a sealed room uses a rulebook to convert incoming Chinese characters into appropriate output characters.

02. Perfect Output: To native speakers outside, the answers are completely indistinguishable from an authentic speaker.

Searle's Conclusion: Syntax is not semantics. Manipulating symbols according to rules does not equal understanding.

SEARLEAN CHALLENGE

The Problem of Commitment & Intent

Case Study: Prompting an AI with *“I promise to assist you tomorrow.”*

- ✕ **No Mind:** System lacks subjective awareness or psychological state.
- ✕ **No Commitment:** System cannot be held morally or socially accountable.

Learning Objective: High linguistic competence does not equate to genuine intentional understanding.

Putting It All Together

The three philosophical pillars of artificial intelligence analysis

FREGE

Representation

Does mathematical geometry in vector space constitute meaning?

WITTGENSTEIN

Use

Does statistical pattern continuation constitute language game mastery?

SEARLE

Understanding

Does contextually appropriate generation constitute intentional mind?

Final Project Assignment

Analyze a contemporary AI architecture through the three philosophical lenses

01

Frege Analysis

What does the system represent internally?

02

Wittgenstein Analysis

How does it learn to use language in context?

03

Searle Analysis

Does it understand, or merely produce appropriate output?

The Big Conclusion

Synthesizing technical progress with philosophical precision

✕ What It Is NOT

A simplistic 1-to-1 equivalence where Frege equals embeddings, Wittgenstein equals transformers, and Searle equals autonomous agents.

✓ What It IS

A conceptual framework where philosophy defines the core questions regarding **Representation**, **Use**, and **Understanding**—while AI provides technical tools to test them.

Q & A / Discussion

Thank you for your time. Let's open the floor to questions and philosophical debate.

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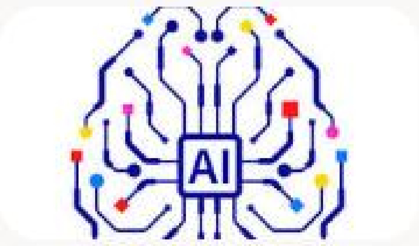

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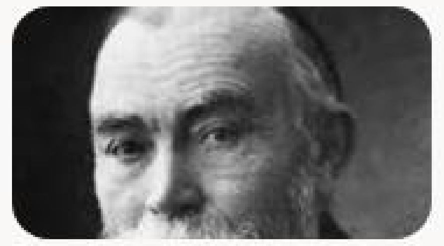

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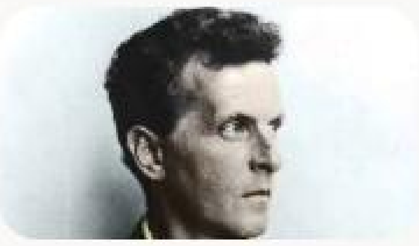

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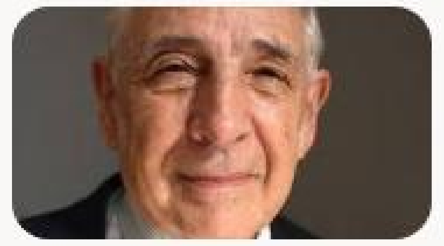

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